

The road to rebuilding confidence

Service Design

Product Design

User Experience

An experience in high school led me to give up my dream of becoming a professional athlete, and now I want to do something for my younger self. Through literature research, I identified my condition as an episode of Functional Movement Disorders (FMDs) and learned that recovery requires comprehensive treatment to rebuild confidence.

Using a systematic service design perspective, I developed a home recovery program for teenage FMDs patients. The core solution features progressive treatment devices, supported by an integrated app for device control, community engagement, and psychological support.



Inspiration

This is something I experienced first hand, in a way this event changed my life, I will always remember it was late autumn 2017...



It was the sports day and I was participating in a 4*100m relay when I suddenly felt a sharp pain in my lower back and then fell down on the track

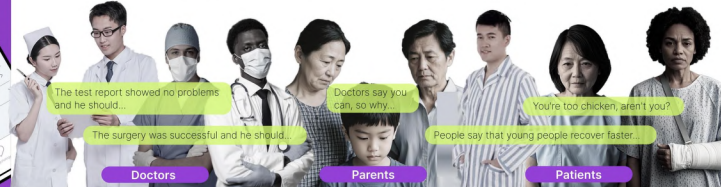


I was quickly taken by ambulance to the hospital, fortunately it was only a small bone pulled from a muscle and required a minor operation.



The day after the operation, I suddenly realised that I couldn't move my leg in any way, it was like I had lost all control of it!

Strangely, medical tests proved that I could lift my legs, so people around me were puzzled.



Eventually I was able to lift my leg, but I ...

Gave up the dream of playing professionally

Becoming less confident in life

Become silent in character

" Later in life I've been trying to do something about myself that day, and after some research I found out that it's actually a mental disorder called **FMDs (Functional Movement Disorders)** . "



Literature Research

In conjunction with personal experience, this project focuses on **adolescent patients**.



FMDs (Functional Movement Disorders)

Simply put, FMDs are psychogenic movement disorders with **psychological factors** as the central trigger, but physical function tests are normal



My situation

In cases like mine, which start with a sports injury and are actually caused by a combination of psychological factors (fear, low self-esteem, worry, agitation...) athletic disorders are also in its scope.

Clinical Data 'FMDs are not rare among adolescents, with a gradually rising prevalence.'

Analysis of FMDs patient data from 1968 - 2019 showed **6.8%** were adolescents. Other studies reported FMDs patients could make up around **10 - 20%** of movement disorder cases in paediatric neurology outpatient clinics.



Most common clinical manifestations of FMDs in childhood and adolescence

I suffer from lower limb muscle weakness as a result of this clinical presentation, and this project will focus on this presentation as well.

*It can be detected with the Hoover sign, a classic sign for diagnosing functional weakness of the lower limbs

Clinical diagnosis requires caution to differentiate from similar diseases.

Factitious disorder

Autonomous symptoms with the aim of hypothetical illness.

Parkinson's disease

Imaging detects reduced dopamine transporter function in the brain.

Malingering

Deliberate display of symptoms for financial and other purposes.

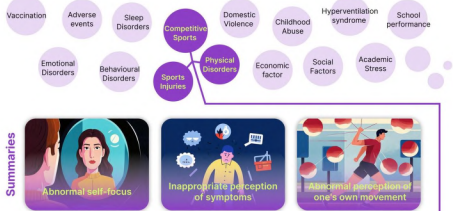
Distinguishing features - Sudden onset and abrupt cessation

'While most movement disorder disorders progress slowly, symptoms in FMDs occur suddenly, can be triggered by events such as trauma or illness, and can dramatically remit, or fluctuate in symptoms during the course of the disease.'

Disease characteristics

FMDs

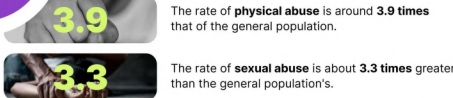
There are numerous triggers associated with adolescent patients with FMDs Etiology



The project focuses on such triggers (Personal Experience)

Sports injury type of causation + Combined psychological factors → FMDs

Research data reveals patients usually have a history of childhood trauma.



User profile

Basic information

Boy 16 years old Student Quiet and a bit withdrawn Medium grades

Household

- Family of 3
- Household disposable income £4W
- Academically minded and strict tutors
- Small obstacles in parent-child communication

Social

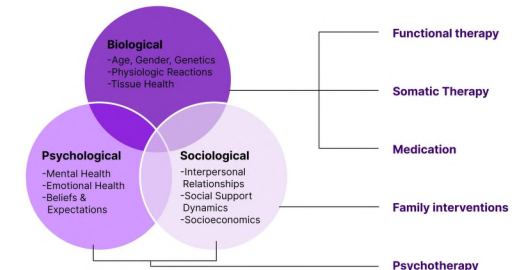
- Friendly and courteous
- Lots of casual friends, few close ones
- Defensive and not very trusting of others

Patient (User) studies

Theory & treatment The current consensus is to adopt a multidisciplinary model for treating this disease as its pathogenesis adheres to the BSP model.

BPS model (biological - psychological - social model) → Systematic service design

The traditional biomedical view deems biological indicators as the final determinants of disease, while the BSP model embraces the patient, disease, and environment. Physicians thus have to take into account the roles of social and psychological factors in addition to biological ones.



Programme planning

'Many studies have highlighted the challenges of developing standardised treatments for functional movement disorders and the need for individualised treatments tailored to the patient's clinical symptoms.'

'Physiotherapy and biofeedback focusing on progressive exercise training is helpful for most patients'

Plan A

Assisted → Developing Confidence → Be Yourself → Increase Confidence

Autonomous

Resistance

Regularity and greater patient acceptance ✓

Plan B

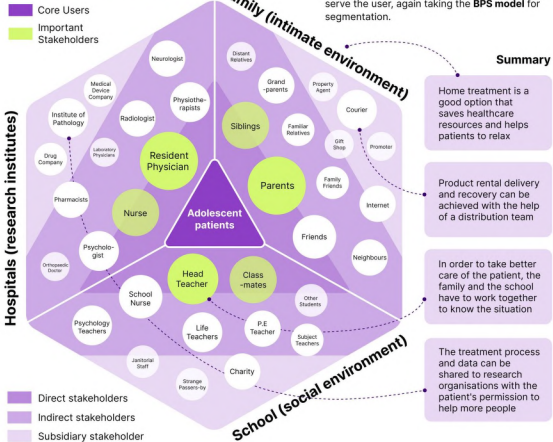
100% of the boost is shown as being used

Actual use of 50% or less

Informs the user of greater satisfaction after completing the goal.

Psychogenic diseases that deceive users can lead to serious consequences ✗

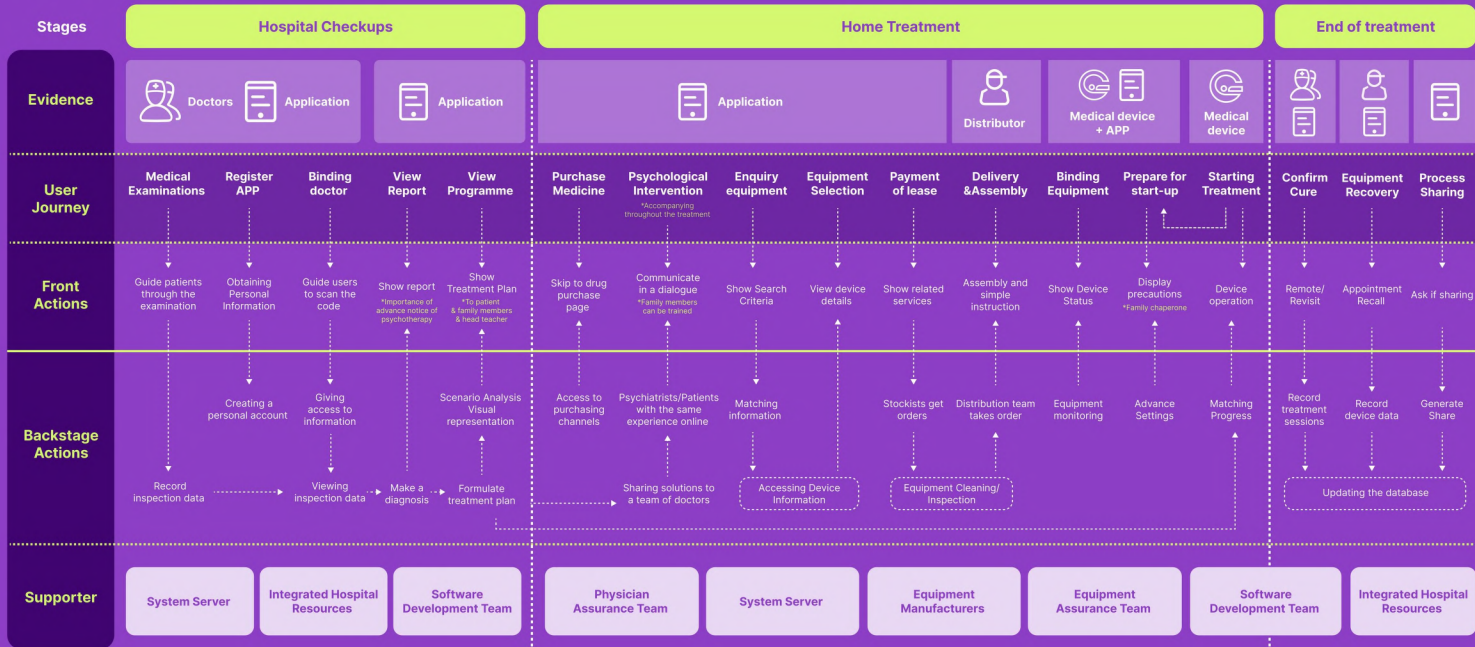
Stakeholders



Definition

Overview	Products	Scenario	Hospitals
	Micro-control hand and foot assistive mobility devices		Pre-programming and preparation
User	Application	Reason	Home
	01 Product Rental and Ancillary Uses 02 Psychological interventions 03 Therapeutic communication between doctor and patient		Medical Products + App Assisted Therapy
User	Time	Time	Time
			Treatment (post-operative) recovery phase, once a day

Service Blueprint



Application
Demo

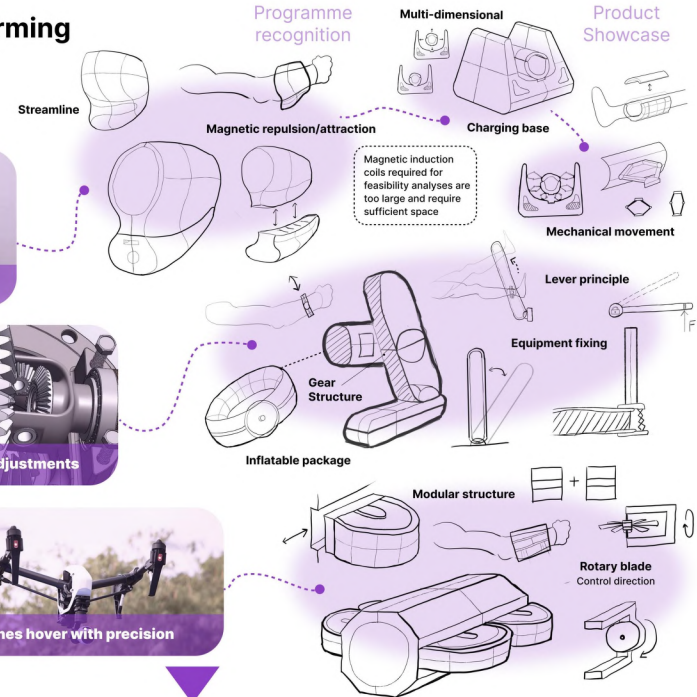
Interface
Testing

Logical plan

Design Process

Demonstrate the design process from both the hardware product and the application software.

Brainstorming



Navigation bar
setting

Status

User Tasks

Core Functions

A No equipment

Search for a shop
Select Product
Payment
Check Logistics

Gain Equipment ●

B Getting equipment

Binding Devices
View Device Status
Tutorials
Getting Started

Start Treatments ●

C Persistence

View Treatment Options
Psychological Counselling
Individual Settings

Psychological care ● Healthcare ● Me ●

UI frameworks



Electromagnet technology

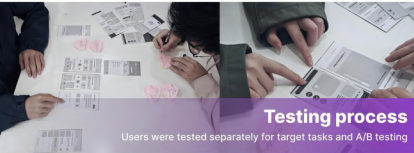
01

Differential Gear Adjustments

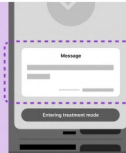
02

Drones hover with precision

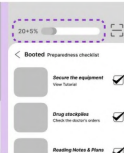
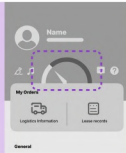
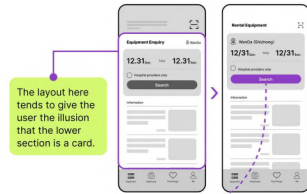
Progressive
therapy ★

**Bright spots****Societal**

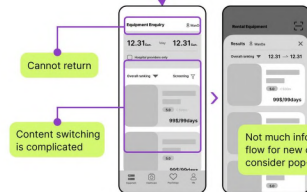
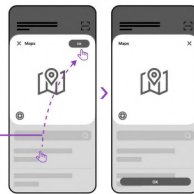
You will see the message left by the previous user when you bind for the first time.

**Psychology**

You can clearly see the current progress after the start of the treatment.

**Interface modification**

The user's finger movements during use are large and can be simplified.



Buttons are similar and may cause mis-touching

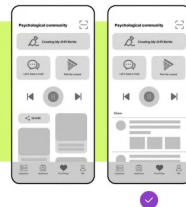
Not much information flow for new content, consider pop-ups

A/B Test**Project 01**

Manifestations of users' shared experiences

Reason

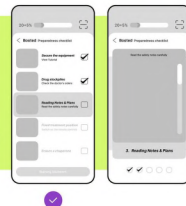
The information on the selected page is more accessible to the user rather than requiring the user to open an additional page.

**Project 02**

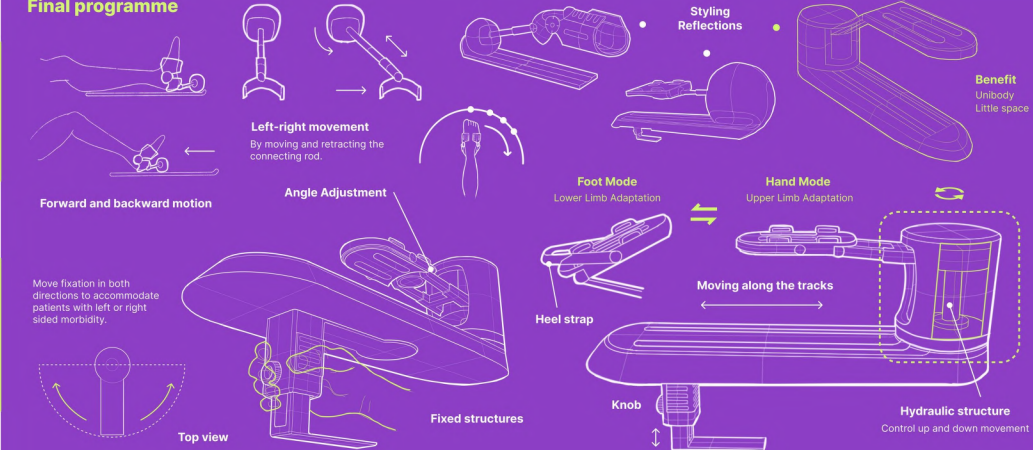
Forms of pre-treatment itemised validation

Reason

Information on selected pages is clearer, non-selected pages require the user to swipe in two directions, which can lead to mis-touching.

**Programme evaluation**

Talking about the background of the project and brainstorming for suggestions

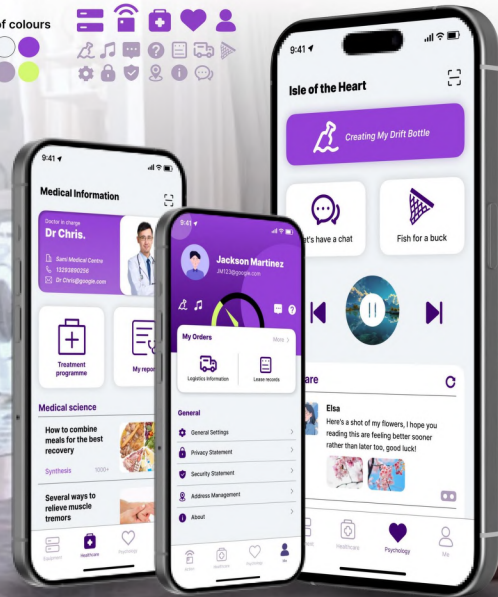
Final programme

Products/Applications Showcase

Use of fonts
Inter

Use of colours

Use of icons

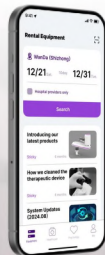


Healthcare

Me

Psychology

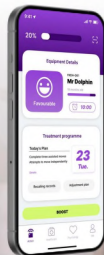
Search page



Detail page



Equipment page



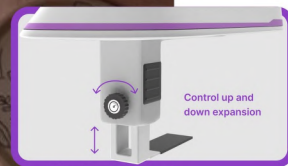
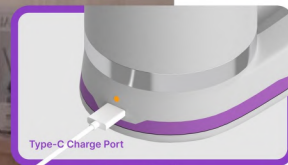
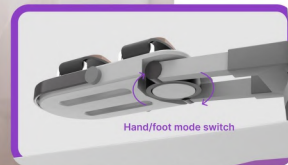
Millimetre control of the mechanics, combined with a three-stage therapeutic logic (Plan A), serves to rebuild confidence (cognition).



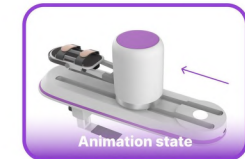
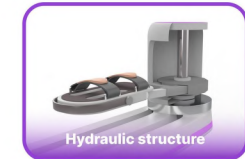
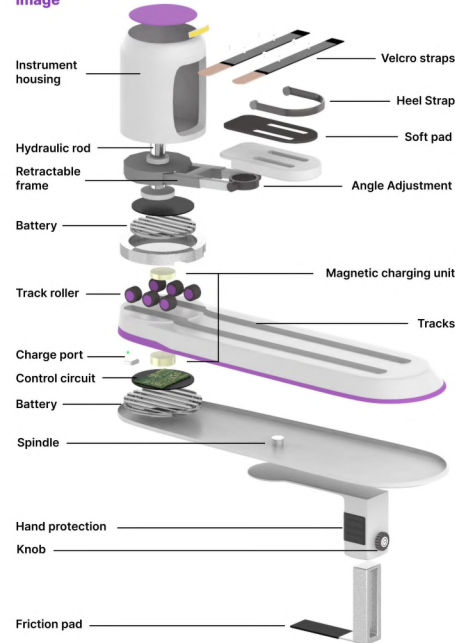
Prelude Preparation Page

Healing state

Therapeutic Manipulation Page



Exploded image



Low-cost accommodation for university students

Economically Sustainable Design

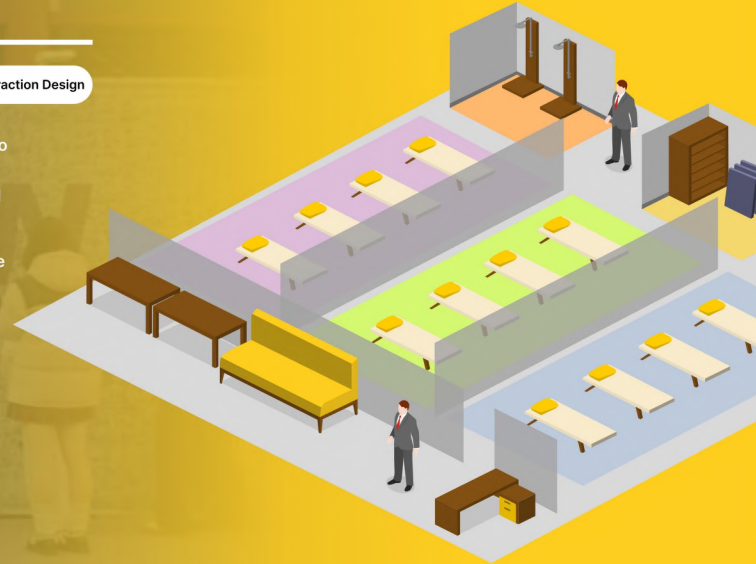
Service Design

User Research

Interaction Design

News of college students' night cycling tours in Kaifeng inspired me to study this travel group. I identified accommodation costs as their key challenge and and, considering the post-pandemic context, designed a government-led space renovation and reutilization plan.

Through a research process of interviews, surveys, and site visits, the cultural center was ultimately chosen as the setting for the solution. The plan was refined through product design, app development, and space planning to ease operational pressures, with the final solution presented as a service programme.



Zhengzhou

Kaifeng

Inspiration

In autumn 2024, a night cycling boom from Zhengzhou to Kaifeng, Henan Province, China, emerged. The daily peak participation exceeded 100,000, boosting Kaifeng's tourist numbers by 30% and stimulating the local consumer market.

It started with a few university students cycling to Kaifeng to eat steamed dumpling, which in turn triggered emulation.

University students first reported Zibo affordable prices

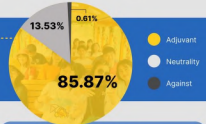
University Students Shoot Vlog Introducing Tianshui

University students' social media agreement to climb Mt Wugong

Looking back at the cases of travel explosion in recent years, university student have almost always been the earliest initiators and responders

University students - a new force in the tourism market

After three years of turmoil during the epidemic era, more and more university students are eager to travel to experience a sense of freedom and enjoy different landscapes.



Short period

85.96%

Choose to go away for two days on the weekend without delaying Monday's classes

Short distance

78.54%

Choose a place that is accessible by transport within three hours of the school perimeter.

Save money

82.57%

Respondents indicated that they would like to keep the overall cost within 500 RMB

"Accommodation is not a core need but it is the most expensive!"

Through observation and communication, I realised that everyone's biggest spend is on accommodation, and this reflects a paradox



Bringing prosperity to the tourist market



Attract more university students



Reduced accommodation prices

*The data are all from CCTV Finance's research questionnaire

Background

On the other hand, the formation of the tourism explosion can not be separated from the strong support and investment of the government



Politics

Promoting the opening of government premises for the convenience of the population.

Economics

Governments at all levels push tourism to stimulate consumption.

Social

In the post-epidemic era, people are becoming more sensible in their spending and investing.

Technology

Innovations in distribution methods have led to a more developed and convenient logistics industry.



How to reduce the cost of living for university students through a form of government-led spatial adaptation?

Research

Interviews with student travellers summarized living space considerations, and prioritise the different dimensions through questionnaires

Qualitative research

Basic tools

(table, chairs, kettle, etc.)

Bathing facilities

Washroom

Power supply capacity

(proximity/number of power sources)

Security and safety check

Privacy

(individual separate sheltered space)

Air conditioning

(temperature and humidity)

Price

Emergency service

(front desk)

Tidiness

Freedom

(personal space size, etc.)

WIFI service

Novelty

(special location/cultural atmosphere, etc.)

13 Dimensions

The order of the list is randomised



Quantitative research

01 - Get more basic information

Can you tell me your current identity? Age? Gender?

Current students or within two years of graduation are considered.

02 - Access to travel-related information

Frequency of trips
(Boundary of one semester)

☐ 1-2 times ☐ 3-5 times ☐ >5 times

Monthly disposable income (Unit: CNY)
(For students who are working, this is after board and lodging)

☐ <1000 ☐ 1000-3000 ☐ 3000-5000 ☒ >5000

03 - Setting up error-proof issues

Maths calculation questions

(The correct result for 36 + 47 is ?)

Excluded data - Crude exclusion of unreliable samples

Filters who don't have the patience to calculate this topic may not be serious about the ordering of dimensions.

Samples with too much disposable income do not fit the scenarios explored in this theme.

04 - Prioritise the dimensions

01 Randomise the 13 dimensions

02 Ask users to rank them according to their level of concern

Scoring rules

The weights are determined by the position in which the options are ranked. A total of 13 options are ranked, the first position is given a weight of 13, the second position is given a weight of 12, and so on.

*Average composite score = (Σ frequency × weight) / number of times this question was completed

05 - Learn More

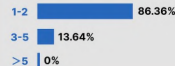
What other factors would you consider?

Results statistics

*Only important data are listed

Frequency of trips

Boundary of one semester



Combined with the fact that the vast majority of respondents had disposable incomes of less than RMB 3,000, it is possible to predict that lowering the price of accommodation is also highly likely to increase the frequency of university student travel.

Learn More

14 respondents gave other factors, of which 6 noted that they would focus on location

A good location generally means a higher price

Shift in thinking

- 01 Getting to a good location made easy
- 02 Transforming spaces in good locations

Prioritisation of dimensions

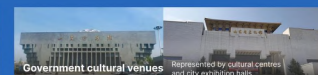
Price	8.05
Security and safety check	7.73
Washroom	7.45
Tidiness	7.14
Privacy	6.50
Bathing facilities	5.18
Basic tools	5.18
...	

Take the top five ratings as the core dimensions, the sixth and seventh are not too far apart and are prioritised as additions to the design.

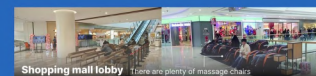
Field visit

Explore potential new government-related living space programmes

Public place - Under the direct or indirect jurisdiction of the Government



Private premises - Potential programmes for government regulation/government facilitation



Note

Ideally, modest modifications that make full use of the current facilities will make it possible to attract university students while at the same time generating economic income for the supply side.

Selected Programmes

In order to better screen the programmes, I talked to some people who regularly communicate with the Government to ask for their views.



Ms Gong
Supervisor
Private Education Organisation



Mr Pang
Public Relations Manager
Manufacturing Company



Ms Kang
Head of Sales
Food Company

"The government will not open up places where there is risk and confidentiality, such as the Public Security Bureau, the Records Bureau, etc."

"Programmes should be easy to administer, have some penalties and be conducive to supply-side acceptance"

"The probability of opening up the non-direct-to-consumer private sector is unlikely, and the comparative risks outweigh the benefits."

"It should naturally be the most feasible to generate revenue without affecting the normal opening and operation of the business."

"Putting yourself in the other person's shoes is the only way to make both parties happy."



01 Removal of programmes that may pose greater risks to government management

02 Removal of programmes that create greater financial risk and expense for providers



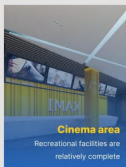
Volunteer Post
Volunteering in exchange for accommodation



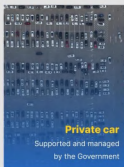
Cultural venues
Opening up parts of the hall for ease of management



Bus stops
Buses that stop at night can be rented in single rooms



Cinema area
Recreational facilities are relatively complete



Private car
Supported and managed by the Government

Strategic decision

Return questionnaires to university students to match scores on **prioritised dimensions and spatial scenarios**

*The price dimension has been removed because it is difficult for respondents to evaluate when there is no cost.

Security and safety check Lavatories Tidiness Privacy Residential Intentions



Very bad



Poor



Fair



Good

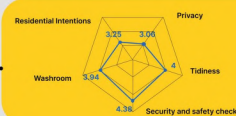


Very good

Five levels were set for each dimension to be judged, using the previous calculations to arrive at a composite score.

Results

Volunteer Post	2.59
Cultural venues	3.73
Bus stops	2.93
Cinema area	2.85
Private car	3.30

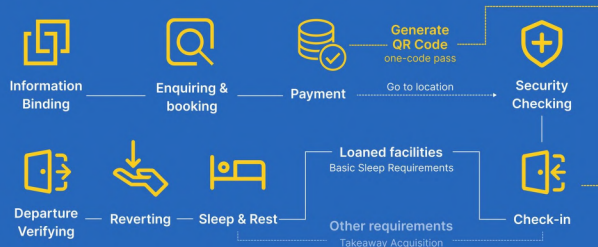


Higher security and tidiness ratings attest to **trust** in sites **managed directly** by the **government**.
Privacy is a priority in design planning.

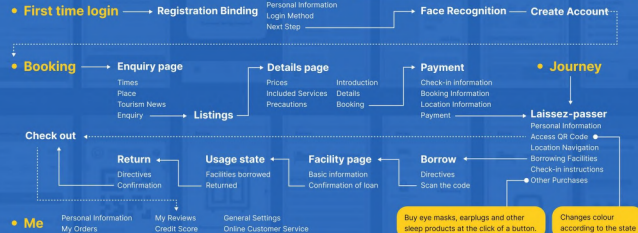
Design Process

Provide the lowest possible operating costs while delivering provide a satisfactory service

User Flow Controlling inputs and improving services

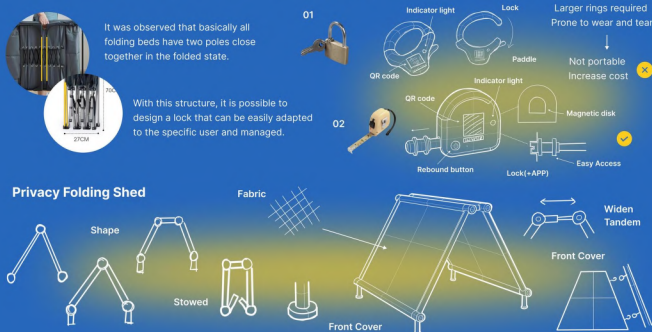


Software design



Hardware design

In order to reduce costs, it is recommended to **use single folding beds** directly on the market.



Service Blueprint

Check in & Rest

Departure

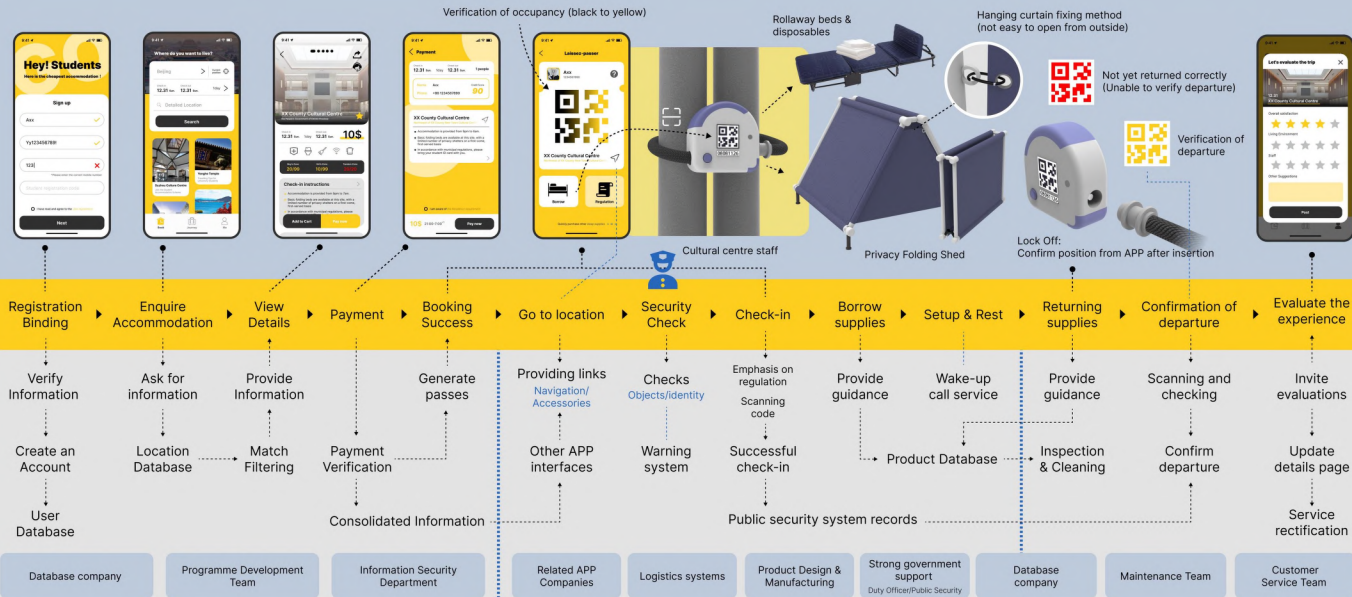
Evidence

User Journey

Front Actions

Backstage Actions

Supporter



Price Settings

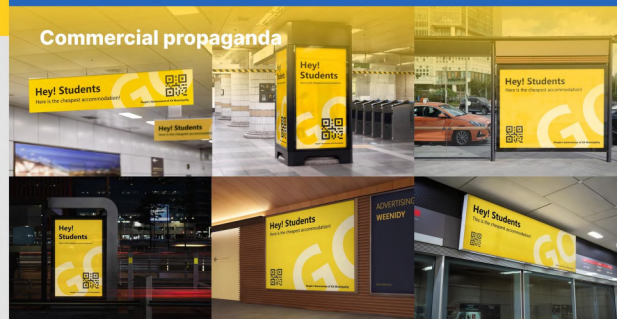
Step price according to demand

Base price —————
Overnight fee
(rollaway bed included)

Extra service	Other need
Privacy board, bathroom facilities, etc	Take-out purchase

*University students with a high enough credit score can rent without a deposit.

Commercial propaganda



Experience = Treasure

Product design

AR interface design

Business design

Inspired by my interest in local history and a film called *The Internship*, I developed a project where seniors contribute to local chronicles, relieving depression and addressing financial needs. It is a pathway linking personal experiences to local history, enabling them to monetize their knowledge.

The final solution, inspired by *active aging* and technological trends, is an AR app that uses AI to convert oral histories into visuals, creating diverse formats for presenting local history.



Inspiration



Local Chronicles's Problems

Interesting details like this make me interested in local Chronicles, but there are many problems that make it hard to find resourceful local history nowadays.



Social Dimension

Failure to form good social connections



Facility Dimension

Inadequate construction of working institutions & collections



Talent Dimension

Difficulty in data collection. Weak cultural identity



Plenty of time



Cultural identity



Personal experience



Well-informed

Just like the film's main theme.

'Experience never goes out of fashion'

Investigation

Oral history

In fact, this method of historical research, which collects the oral memories and personal experiences of individuals or groups of individuals, is of great value to historical research and is called 'oral history'.



Oral history

What benefits can it bring?



Observation

By observing my grandparents, their friends and my elderly neighbours, I found that an interesting phenomenon is that today's elderly people **spend a lot of time on their mobile phones every day**.

Relevant Data

As of June 2023, the size of China's Internet user group aged 60 and above had reached **140 million**.
— From CNNIC

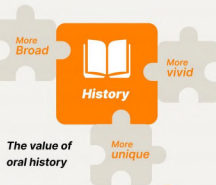
Nearly **10%** of elderly people go online for more than **6 hours** a day.
— From Tencent and People's Daily

As the Tencent report writes
"More than 100,000 older people across the country are showing extreme loneliness online."

Phenomenon
Internet addiction



Nature
Mental disorder



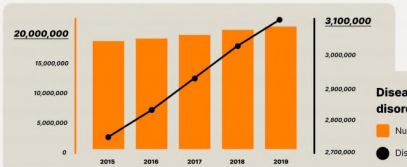
Analysis

World

Globally, more than **14.1%** of persons over 60 years of age suffer from psychological disorders.

Depression is among the most common mental health problems, with a global population of depressed older people of up to **80,000,000**.
Approximately equal to the entire population of Germany

China



Disease burden of depressive disorders in the elderly in China

Number of people with the disease
Disability-adjusted life years

27.2%

Globally, about a quarter of all suicides occur among people aged 60 and over.

23%

China's prevalence of depression in old age is 23% globally, and even reached 26% (2019).



Academics often analyse the causes of psychological problems in older people from a number of perspectives



Inadequate social participation



Participation of older persons in the compilation of local histories to enhance social participation.

Life Review Theory (Butler 1963)

It helps the elderly address unresolved issues from earlier in life, alleviates anxiety, and effectively relieves depression. And it is widely recognized abroad.



Lack of care for the family



Many elderly people and their families lack awareness of mental health issues, so proactive care and support are essential.



Degradation of body functions

Negative ageing → 1990s → Active ageing
Influenced by Positive psychology and Postmodern thinking

Appeal to the elderly themselves not to be filled with despair because they are no longer young.

Calls on young people not to neglect older people just because they are no longer young.



Enlightenment

Designers need to address ageing with a positive mindset and can try to use bold and innovative approaches.

User research

Retirement is an important point for the shift in the level of social participation of older people. My grandparents did the same.

At first, They made some attempts to adjust to retirement.

Information	Attempt	Result
 My grandma Age: 74 Occupation: Textile factory worker	 Cross Stitch	 Give Up Exacerbation of visual impairment Leads to lumbar muscle strain
 My grandpa Age: 73 Occupation: Beer company manager	 Invest in the food industry	 Give Up Consuming a lot of energy Disproportionate returns

User Interviews

Interviewer	Interviewee
- What do you get out of it? - Any regrets? - Will you adapt to new technology? - Other suggestions?	- Economic benefits Government Recognition - Lost friends Old House - I will learn and keep the tradition. - Don't get tired of using it.
Summary	- People tend to look for opportunities in areas they are familiar with after retirement. - Currently there is a lack of a home-based and healthy model of social participation for retirees. - Ideally, design a business cycle that can generate some revenue for older people. - Older adults adapt to technology but maintain long-held habits.

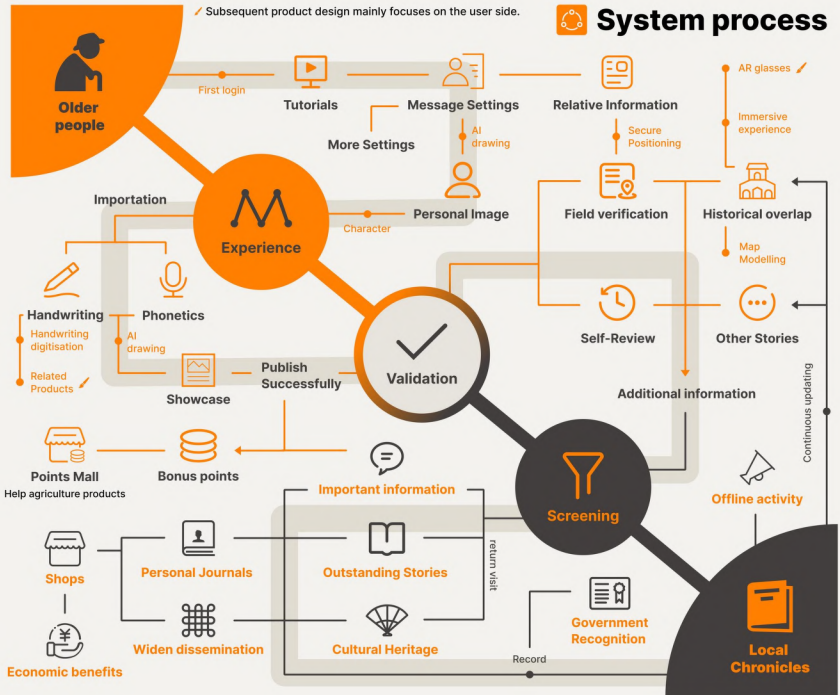
Definition

- Products that connect **local chronicles** with the spiritual enrichment of **older adults** through the **Life reviews theory** and the **Oral history** while bringing additional value to the user.



Business Analysis

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
- Equipment manufacturer - Government organisations - Map Development Companies - Publishers and distributors - Agricultural dealers - Historical researchers	- Collecting experiences and verifying them - Extract core into revenue	By collecting the experiences of older people, it is possible to enrich history and bring many benefits to older people, while at the same time turning information into revenue and also creating a sharing platform for older people.	- Make a return visit - Organise offline events - Personalised coaching	- Recently Retired Senior Citizens - Users who are not comfortable going out - Children of the elderly - History buffs
Key Resources		Channels		
- Creative forms - Wide range of information - Designed for the Elderly		- Social networking platform - Sensitisation of children - Government Co-operation		
Cost Structure		Revenue Streams		
- Equipment R&D - Production Costs - Labour costs - User incentives - Advertisement Costs - Legal protection		- Advertising revenue - Sale of information - Sales commission - Government Subsidies		

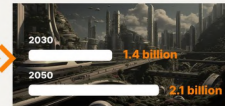


System process

Feasibility analysis

- As the current technology is not perfectly adapted, the technology used in this product is based on the current development trend of short-term future applications, with a certain degree of foresight and feasibility.

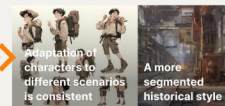
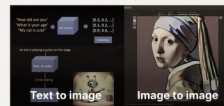
Elderly population



AR glasses



AI drawing



Handwriting digitisation



Map Modelling



Products

Hardware Products List

01 Pen (Handle)

02 Electronic handwriting book

03 AR glasses

04 Storage box

01



Pen end as handle

Keypad area

- Voice button
- AI key + scroll wheel
- Back button



Start button & Shutter

Sound location

03



Magnetic pin charging



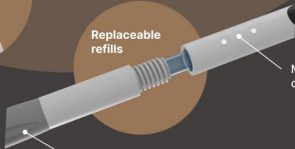
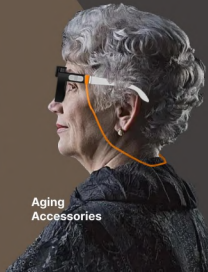
Projector



Surveillance camera



Focus & volume adjustment



Replaceable refills

Rubber grooves

Limit the most appropriate direction for holding the pen

04

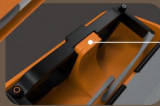


Storage box

Computational unit
Detachable for travelling



Type-C powered



Glasses charging stand



Effortless



Save effort

Observe the human-machine relationship and set the position of the indicator ray emission at an oblique angle of 45 degrees.

Electronic Handwriting Book

02

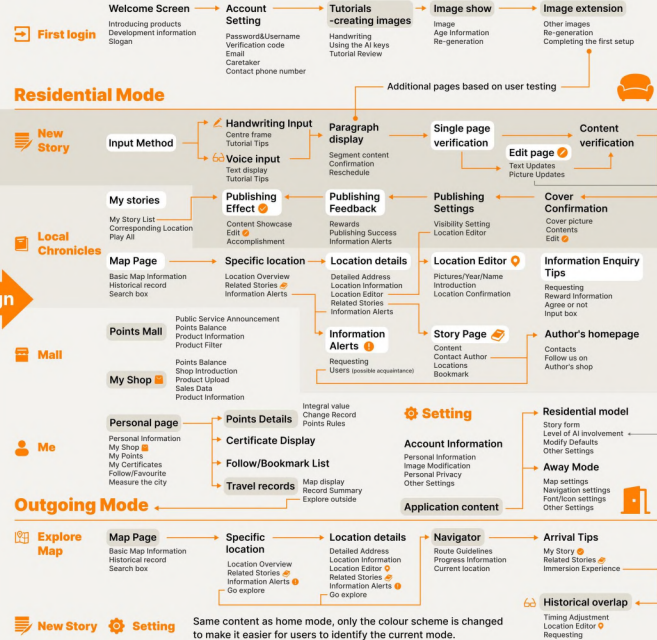


Electronic Handwriting Book

Directional sign

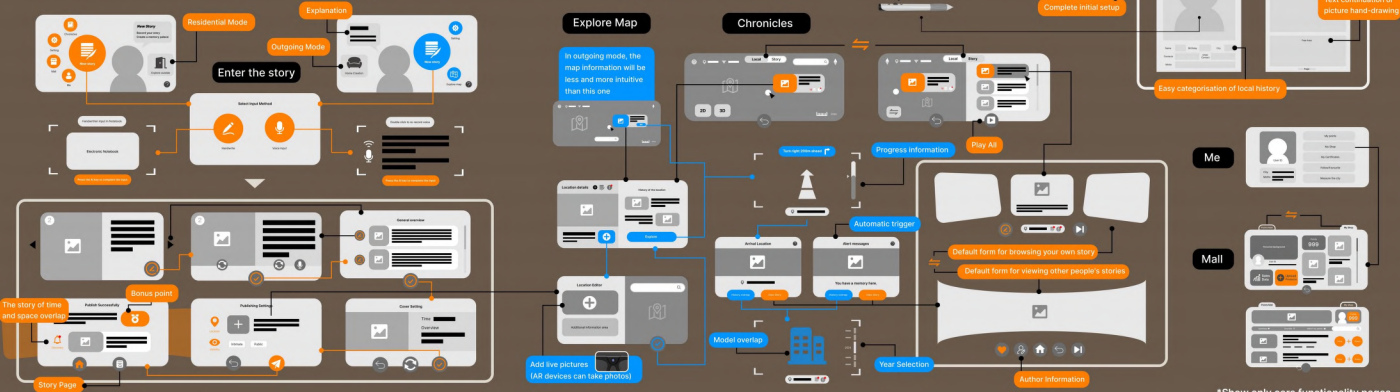
UI Framework

*Note: Since some of the links between the function blocks will cause overlapping, which will lead to logical confusion, the same icon will be used as a link in this case.



Interfaces

AR interfaces



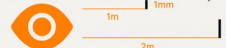
Develop norms

Principle

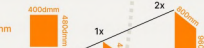
dmm

Distance-Independent Millimeter

The Meta team developed the 'dmm' concept, which forms the underlying specification for the UI design of XR products.

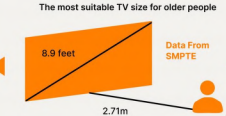


Both of the above cases correspond to 1dmm, which allows for consistency of content across different distances and screen sizes.



The same principle applies when expanding to the screen.

- Workflow**
- Test the most suitable screen distance
 - Extrapolation of interface size by combining the dmm principle and known TV data
 - Test the smallest acceptable text size and icon size
- *Combined with Meta recommended sizes



This translates to a television set with a length and width of approximately 236cm and 133cm.

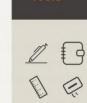


The size of the interface can be tailored to the TV viewing habits of older people.

Testing



Tools



Results

Consistent with the Oculus team's test results!

Distance 1.0m

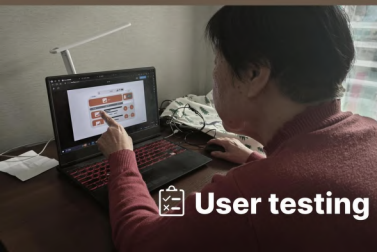
236cm = 2.71m

Interface size 87×49cm

Text size 25dmm

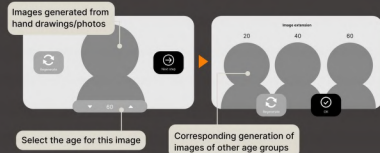
Icon size 50×50dmm

*The result is the minimum size of the design, which is used as the bottom line for the actual design.



User testing

01 Users want to use images of themselves at different ages in story presentations



02 Unsatisfied with story segmentation content



03 Can't find how to switch between the two malls



Showcase

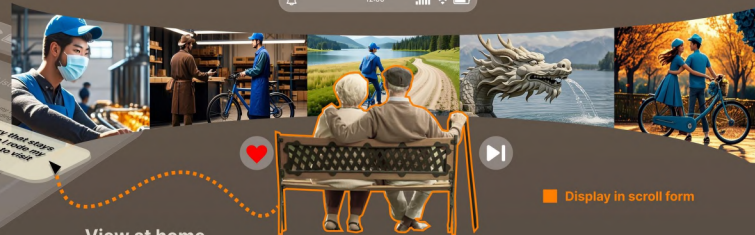
Homepage Residential Mode

Homepage Outgoing Mode

12:00



Setting image



View at home

Display in scroll form

In order to prevent vertigo in the elderly, the focus during this type of presentation is centred on the back of the head, in line with the principles of retinal imaging.

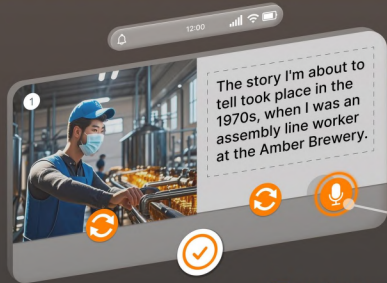
Initial setup

Enter the story

Field exploration

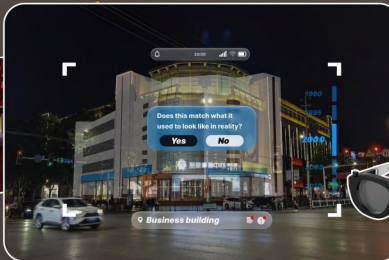
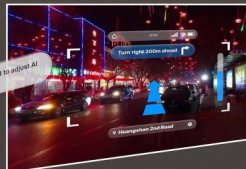
Economic values

Historic value



Edit page

Navigator



Arrival Tips

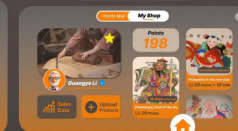


Historical overlap



Online Mall

According to the point system, people can redeem the aid products. At the same time, for some intangible cultural heritage can be sold to promote culture. For ordinary users, they can sell their stories in the form of personal journals.



Local Chronicles

When the information is enriched in large quantities, the local history can be presented in a variety of forms.



In addition to the selected buttons being spatially drawn closer together, a prompt message floats on the buttons of the handle.

Auxiliary operations

Simulation- Key to Victory

Interaction Design

Product Design

User experience

A key loss during my school team career drove me to investigate its cause. Surprisingly, our opponents trained less than we did, suggesting a key to improving efficiency. Interviews revealed simulation training as a powerful tool, inspiring me to analyze its elements and innovate.

Using Harris chart analysis, I designed a tiered-response electronic headband that alerts players to on-field scenarios, boosting mental resilience and reaction speed while tailoring training for different positions.



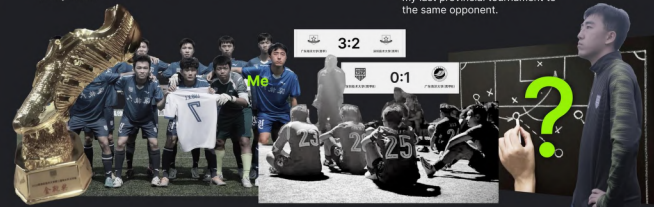
Inspiration

I In my freshman year, winning the **Golden Boot** award in the freshman game, I made the varsity team.

II I have participated in **various competitions** at provincial and municipal levels with the school team.

III A couple of heartbreaking **losses** in **key tournaments**, particularly my last provincial tournament to the same opponent.

IV I'd like to know what **exactly** causes people to underperform in key games.



My group is a **general student group** (no professional training experience)



There's **not much difference** in the level of either the players or the coaches.



I think they may have had **more practice time** together and so performed better.



Surprisingly I've found a couple of strong teams that **train less than we do**.

FEWER
TRAININGS



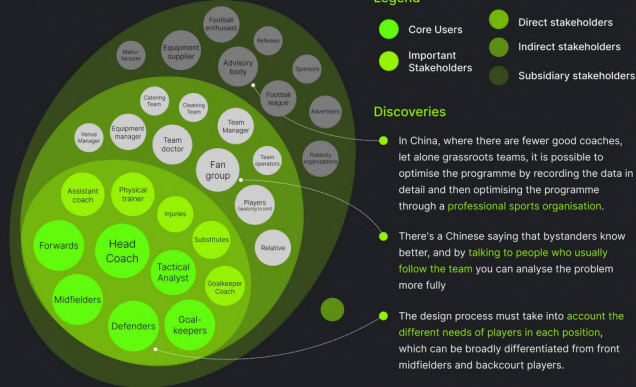
BETTER
OUTCOMES

What are the **key factors** that balance the scales?

Research

Primary research is primary, secondary research is supportive

Stakeholder



User Interviews

What do you think is the problem with the current team's training inefficiencies?

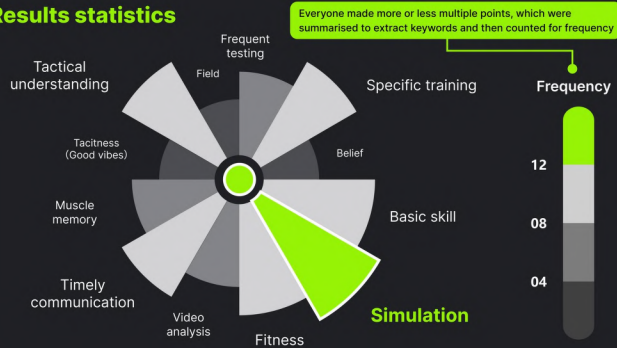
Which training do you feel has been effective in improving your performance during a race?

For players and coaches

For fans and staff member

Through observation and knowledge, which part of the training do you think is the most important?

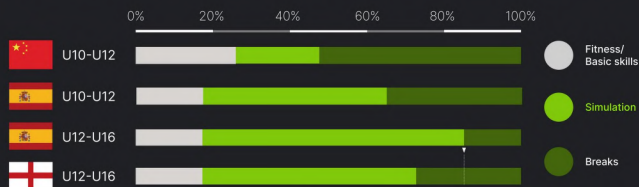
Results statistics



Information analysis

*The structure of elite youth football training sessions

*Roca & Ford, 2020. Decision-making practice during coaching sessions in elite youth football across European countries. Science and Medicine in Football.
Feng et al., 2023. Comparison of training activities and coaching behaviours in youth football coaches from Spain and China: a case study. International Journal of Performance Analysis in sport



The comparison reveals the importance football powerhouses place on **simulation training**, which to some extent supports our interview findings.

These two teams just happened to be the opposing sides in the Euro 24 final, so perhaps a little difference in simulation training as a teenager played a bit of a role

Elemental analysis-Simulation

Match scenarios

The desire for championships and victories fuelled the players

Give it all you've got. After all, no one wants to lose

High-intensity environments are more likely to be out of control

Your opponent will never give you time to think

The atmosphere built by the fans is exhilarating

For example, the Spanish player Reina would fill up his car with petrol before every match

Players need to adapt to different venues and facilities

Coaches and referees will show a different style than they do in practice

Aims Pursued

Range of motion

Emotional Condition

Decision-making Time

Fan Vibe

Personal Superstition

Football field facilities

Coaches & Referees

Just do what's on the training schedule, lack of enthusiasm

No need to fight hard to avoid hurting teammates

The atmosphere is much more relaxed and there are few emotional outbursts

Players have time to think about their next move

Only team officials and family visit the training ground

Superstition exists but it's not the same as an official game

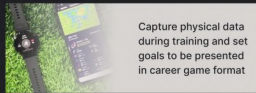
Familiarity with the training ground, facilities, and equipment

Coaching is more patient and the role of referee is difficult to revert to

Initial programme

- 1 Remove the more subjective elements of 'range of motion' and 'personal superstitions' and extend the programme with the other elements.
- 2 Potential scenarios were analysed using the **Harris profile method** and the highest scoring scenarios were developed and designed.

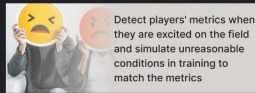
Aims Pursued



Capture physical data during training and set goals to be presented in career game format

	--	-	+	++
↑ Importance				
Practicality				
Innovativeness				
Feasibility				

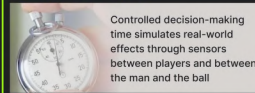
Emotional Condition



Detect players' metrics when they are excited on the field and simulate unreasonable conditions in training to match the metrics

	--	-	+	++
↑ Importance				
Practicality				
Innovativeness				
Feasibility				

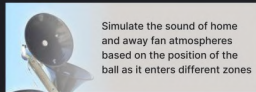
Decision-making time



Controlled decision-making time simulates real-world effects through sensors between players and between the man and the ball

	--	-	+	++
↑ Importance				
Practicality				
Innovativeness				
Feasibility				

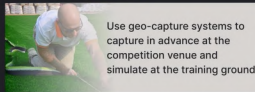
Fan Vibe



Simulate the sound of home and away fan atmospheres based on the position of the ball as it enters different zones

	--	-	+	++
↑ Importance				
Practicality				
Innovativeness				
Feasibility				

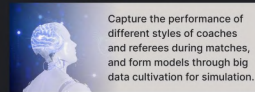
Football field facilities



Use geo-capture systems to capture in advance at the competition venue and simulate at the training ground

	--	-	+	++
↑ Importance				
Practicality				
Innovativeness				
Feasibility				

Coaches & Referees *Mainly referees



Capture the performance of different styles of coaches and referees during matches, and form models through big data cultivation for simulation.

	--	-	+	++
↑ Importance				
Practicality				
Innovativeness				
Feasibility				

Practicality Can it be applied consistently without creating new problems

Innovativeness Whether the programme is innovative and refreshing

Feasibility Availability of current technology and complexity of operation

Refinement



01 Input-Activated Logic



For attacking players - work on the ability to handle the ball quickly with the ball in play (shooting/passing)



Regular Running State



Ball-holding state



Failure to make a timely pass (shot)

*The logic applies to personal training

For defence player - Training defensive complementary skills and defensive positional sense



Regular state of defence



Opponent enters defensive zone



Failure to create a timely disruption

*The logic applies to physical training

For midfielders

Switch the logic around so that being in an area with no defenders triggers the first level of response, which in turn trains the ability to run without the ball.

For personal training/goalkeepers

Adjusting the activation logic to a random response allows for emergent response training and agility training.

02 Outputs - transmission routes

Based on the motivation logic, the product needs a reminder that can be responded to in a **graded manner** and can **do so quickly and without creating a major obstacle** for the player.

Modality of perception	Hearing	Taste	Olfactory	Vision	Touch	Intuition
Clearly graded	████████	██████	██████	████████	██████	██████
Rapid response	████████	██████	██████	████████	██████	██████
No interference	████████	██████	██████	████████	██████	██████

First level response

As the first level of response serves as a catalyst, it can follow the Hong Kong traffic light model of reminding pedestrians to pass quickly by means of a sharp sound.

Secondary Response

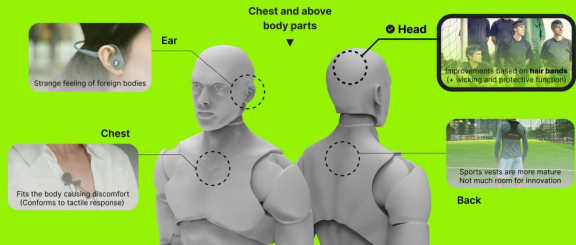
For better results, the secondary response uses touch as the sense of perception, alerting the player through vibration or tightening.

Adaptable parts

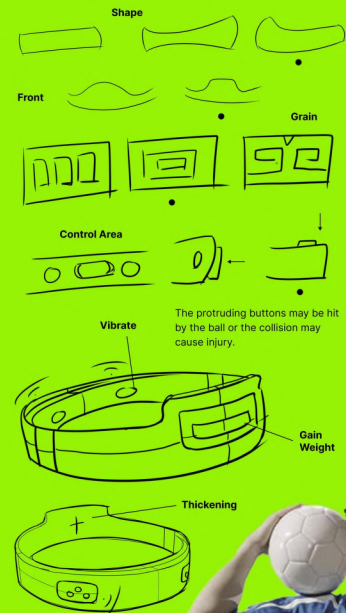
Use current player equipment as an entry point to explore alternative or improved designs to reduce player discomfort and enhance simulation training



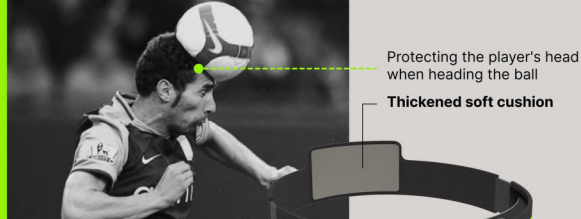
Perceptual modalities corresponding to the different responses, firstly excluding scenarios that are far from the ear



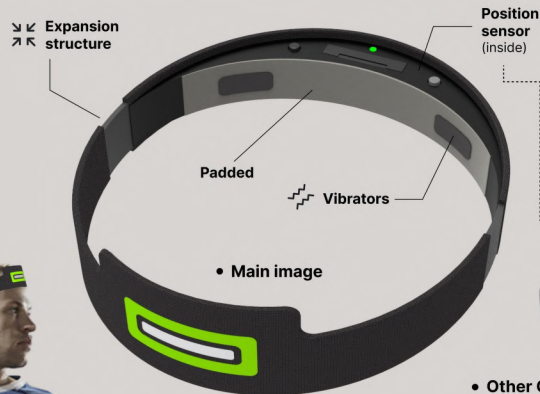
Design Showcase



Sketch

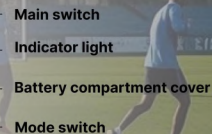
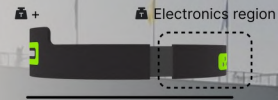


Speakers



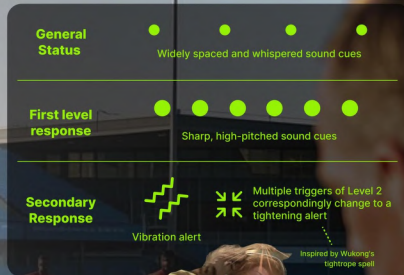
Product Effect

Front elevation view



Real performance
Take attacking player as an example

Other Colourways
Easy to differentiate during simulation training



Coaches can analyse each player's ball handling speed and make adjustments, as well as set up rewards and penalties based on performance.

Keeping your jewellery organised

Product Design

Project Management

Business Design

This project was driven by the company's strategy to enter the jewelry storage market with a mid-range product. As the product lead and ID designer, I defined the PRD by revisiting an abandoned concept and incorporating insights from competitor analysis.

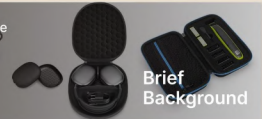
The design optimized storage size and reduced logistics costs through detailed adjustments. After refining the prototype with suppliers and addressing testing issues, the product launched successfully during the peak season, earning strong user feedback.

*Some information has been pseudonymised and hidden due to commercial content



Project Origins

The team started as a bag maker, and the current main line is to design and produce **storage (protection) products for high unit-price merchandise**, mainly for overseas markets such as Europe and the United States.



Brief Background



Users

Users are more likely to choose specialised storage products for merchandise with high unit prices



Interests

Higher-unit-price items corresponding to stowage products tend to have higher premiums and more benefits



Markets

Leverage China's low-cost supply chain and overseas culture of organisation for commercial success

Project Planning

01 Long-term goal - Creating a brand with a foothold in jewellery organizer

*Three-step planning based on development costs



01



02



03

The company has formed a more complete product matrix in the field of electronic equipment storage, in order to expand the market, decided to **enter the field of jewellery organizer**.



→ 02 Current goal - Introducing a jewellery stand that is competitive in the mid-range market ★

This range has a large number of users and most of them are more sensitive to value for money



Mid-range market

Creating a more positive rather than cheap brand image (Chasing the low-grade market)

- The company had carried out a design for a jewellery stand before I joined, but abandoned it because the estimate was too high and risky. **If this product (let's call it a regular model for now) is successful in opening up the market, the previous programme can be fast-tracked to be sold as a combination of Pro models.**

Project Objectives



Propose a product solution with a price advantage and competitive overall indicators.



Timely listing, year-end (Black Friday/Thanksgiving/Christmas) is the peak season for European and American markets.



Finding the right supplier and communicating well to lay the groundwork for continued future co-operation.



Collaborate well across departments and gain relevant development experience



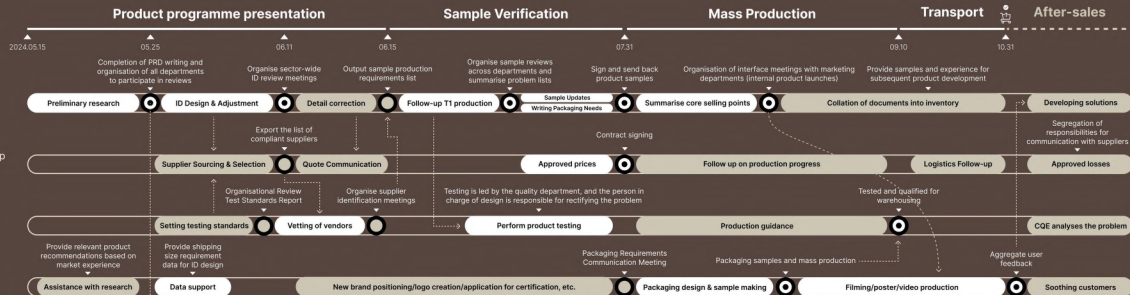
*Due to the project timeline and in order to better understand the product, I (Product development director) was appointed as the ID designer for the regular jewellery stand.

Me
Director & Designer

Business Department
Purchasing & Commercial Follow-up

Quality Department
SQE, DQE, CQE

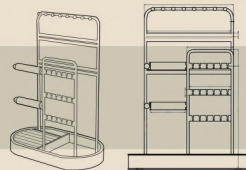
Marketing department
Sales/Operations/Graphic Design



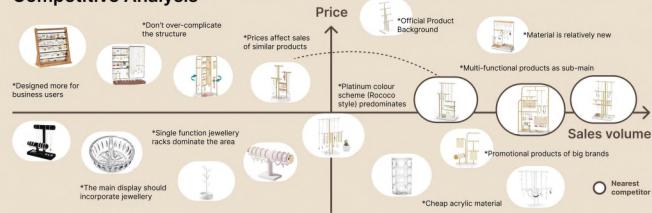
Programme research

Previous programme (Section Pro)

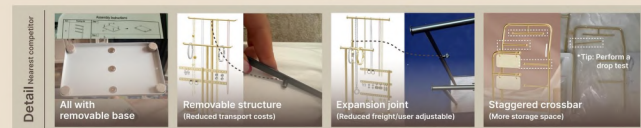
- Framework--- Adopts front and back tiers of sectional jewellery holders
- Material--- The main body is a metal frame. The base is velvet lined with a PU wrapped runway circle



Competitive Analysis



Key Partners			Key Activities			Value Propositions			Customer Relationships			Customer Segments 		
Raw Material/OEM Supplier			Research & Design Supplier Selection Quality Testing Online Sales Logistical transport Mass Production			Cost-effective products Providing users with quality products that make jewellery storage and display well organised Variety of options Timely follow-up of new products on the shelves after the successful launch of the first model (regular version)			Organisation of holiday promotions Quality after-sales service / regular visits Channels Amazon Selling YouTube Promotion Website Operations			Residential customer ● Mostly female users ● Upper middle income ● Have more jewellery or accessories Business customer ● Jewellery sales offline shop ● Jewellery fairs use ● B-end shop wholesale Emerging markets customer The hospitality industry is a growing jewellery storage market, and some shops offer the jewellery organizer product as an in-room amenity		
Logistics and Transport Companies														
Overseas Market Agents			Key Resources											
Overseas Warehouse Management			Cheap and complete supply chain											
Product Certification Organisation			Experienced and systematic team											
Cost Structure						Revenue Streams								
Company Operation		Logistics and Transportation		Platform (advertising/sales) costs		Retail merchandise receipts		B-market sales revenue						
Supplier costs		Overseas Warehousing/Agent		Patent/Certification Application		Intellectual property licence income		Government/User/Platform Incentives						



Product Requirement Document

Multi-functional zoning

It has been suggested that the regular model to focus on specific jewellery storage (e.g., earrings/necklace), but the research led to adopting the multifunctional partition design.

Details - Cost Control

Appropriate detailing can be considered for cost control (creating more marketing space) without compromising functionality and user experience.

Familial Series

In order to create a better family product effect, some of the features of the previous scheme drawings can be retained and adjusted in size to differentiate them

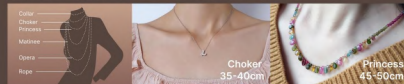
ID Design

Following agile development principles, I (with an undergraduate degree in design) doubled up as the ID designer for this market-occupying (regular) product

Product Size

01 Determine the size requirements needed to meet basic functions

Research shows that the most popular types of necklaces (Classification by length) in Europe and the United States are the Choker and the Princess Necklace.



The drop length is half the total length ($\leq 25\text{cm}$), so set the product height to **27cm**.

Similarly, the height of the bracelet holder is determined to be **10cm**, based on the value of the standard wrist size of the European and American population.



Competitor Observation

By observing the competitors, it can be seen that the outermost hooks are often bumped during delivery, so leave enough space.

Reserve 2.5cm

02 Determine size requirements that can reduce costs based on Amazon FBA shipping standards



FBA-North American region
Carefully graded parcel weight requirements



Comprehensively meet the shipping requirements of the two target markets, set the product size against the Amazon FBA fee schedule.

Height **<4cm**
Weight **<460g**

*Allow for the height and weight of the package when designing the product

Approximate savings for North America

1.1\$ $\approx 8\text{¥}$

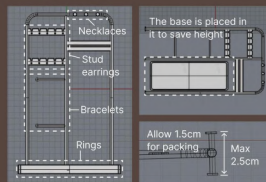
0.5€ $\approx 3.8\text{¥}$

Approximate savings for Europe



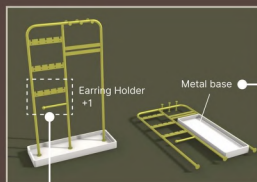
FBA-European region
Carefully graded parcel height requirements

Programme



Assessment

Users tend to have more stud earrings and bracelet holders that are too high increase the risk of damage from drops.



Supplier Selection

Work with SQE to compare and audit suppliers submitted by purchasing staff;
▼ The following table shows only the differences in the comparison results

Comparative term	OEM Supplier Y	OEM Supplier W
Quote	¥X (Single corrugated packaging included)	¥X+10 (Without packaging)
Metal base detail	Non-integrated base (with welded joints)	One-piece base (Moulding required)
Welding detail	Arc welding Smaller weld joints	Larger solder joints Conventional soldering

Sample Production

T1 Material Confirmation



Select Hollow 6mm Tubing



The supplier recommended **MDF material**, so it was requested that separate samples be made in metal and MDF

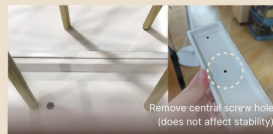


- Lower quality MDF material
- Inset screws for better aesthetics
- Metal base tends to rattle when putting in jewellery (cheap feeling)

Problem modification

01

Assembly tests revealed that there was a risk of misassembly by the user due to the fact that the middle piece of pipe was not in the centre (according to the drawings).



Remove central screw hole (does not affect stability)



02 Supplementary shims



03 Uniform storage of screws

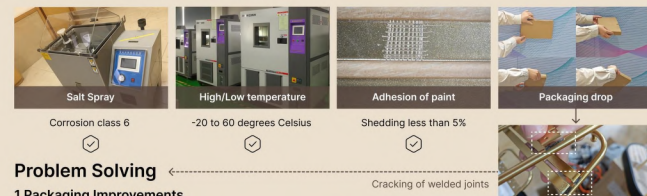
T2 Substantial compliance



Vendors point to staggered design as more realistic

Risk Exposure & Resolution

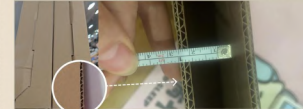
Test Items Assist DQE with test projects



Problem Solving

1.Packaging Improvements

1.1 Single-layer corrugated box to double-layer corrugated box



Increased packaging costs; There is a greater risk that packaging will exceed 4cm, leading to increased transport costs.

1.2 Increase cling film wrapping and tape fixing packaging



Impact on the user's unpacking experience; High labour cost due to complicated operation by workers.

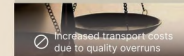
2.Process Improvements

2.1 Conversion to conventional welding (Increased welding area)



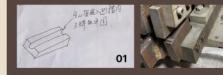
Loss of speciality plus

2.2 Conversion to solid tubing (Batch cost to be borne by supplier)



Increased transport costs due to quality overruns

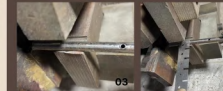
2.3 Tubing embedded in groove and welded



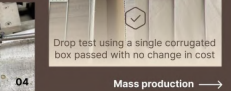
01



02



03



04

Drop test using a single corrugated box passed with no change in cost

Mass production

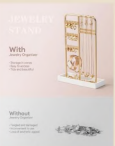
Propaganda Communicate the core selling points of the product through requirement documents, internal conferences and other forms, and the marketing department will carry out the design performance.

Packaging Write package content and requirement documents to marketing department

Product Information	Platform requirements
- Title (Brand/Name) - Core Selling Points - Product Line Drawing (Labelled Dimensions) - Parameter information table - After-sales contact (Website/QR code/email)	- Agent/Agency Contact (EU countries/UK) - Relevant product certificates (CE marking, Triman marking, etc.) - Label 01 (Serial Number, Manufacturer Information) - Label 02 (Shipping information, traceability barcode)



Home Page & Main Image Internal launch to explain the core selling points of the product



Basic storage function



Multi-Area Storage



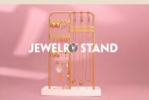
Product size



Exquisite and solid craftsmanship

- Other points**
- Easy to Assemble**
Can provide the user with the relevant links to install the tool
 - Ideal Gift**
Can provide the user with the relevant links to install the tool
 - After-Sales Service**
Timely response to your claim

Promotional Video Marketing department shoots promotional videos (official & invited celebrities)



Interesting detail

The horizontal bar here is designed to **avoid tangling of necklaces on the front and back sides**, while some users find it **suitable for storage and packing** as well. This suggests that users will make more convenient choices based on the actual situation when using the product.

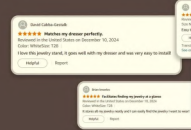
Product Sales After production completion and SQE's qualification confirmation to the warehouse, promptly follow up the logistics and sales data of the sales department.Communicate and follow up with CQE in case of customer complaints.



Previously the products were inspected by SQE and sampling in the warehouse, and then confirmed to be qualified for delivery and shipment, and arrived at overseas warehouses

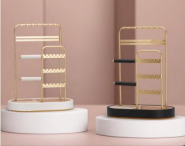


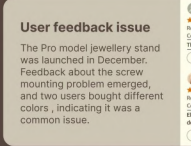
User Reviews As of 2024.12.20, there were no customer complaints for the regular model
Seeing users' favourites and suggestions is the biggest motivation as a product manager



User feedback issue

The Pro model jewellery stand was launched in December. Feedback about the screw mounting problem emerged, and two users bought different colors, indicating it was a common issue.





* I had left my job at the time of the feedback on this issue and I think the possible reasons are:
01 The equipped screws are not long enough, or the supplier's selection of screws is the same as the general model, resulting in a mismatch.
02 The Pro version has a more complex base material, and may not have handled the internal sanding when the holes were opened up.

Sales Stage

Current stage

01 Regular models steal the show; Use links to existing brands.

02 The Pro model was put on the market; More options available.

03 Handling user feedback; Optimising the product.

04 Scale-up of production; Adoption of independent brands.

05 Operate a stand-alone website. Developing serialised products.

Gradual expansion of product sales, brand awareness and market size

Summary



It is very important to get help from the supplier (technician) in the first co-operation, so good co-operation outweighs the first benefits.



Don't limit yourself to just one approach, e.g. design detailing, supplier selection, business communication, etc. can all lead to cost control.



Having multiple items in one link gives users more choice, but there is also a risk that one of the items will be faulty and affect the overall user rating.



Once the super-sellers in the space undercut prices it can have a major impact, so keep innovating and cultivating unique advantages.